Solving the Boundary Layer Equations with a Polynomial Ansatz

Soumadeep Ghosh

Kolkata, India

Abstract

This paper presents a comprehensive analysis of solving the Prandtl boundary layer equations using polynomial ansatz methods. We derive the governing equations from first principles, establish the mathematical framework for polynomial approximations, and present rigorous proofs for the existence and uniqueness of solutions under specified boundary conditions. A systematic algorithm for determining polynomial coefficients is developed, and numerical comparisons with the exact Blasius solution are provided. The cubic polynomial profile $u/U = \frac{3}{2}\eta - \frac{1}{2}\eta^3$ is shown to approximate the boundary layer velocity distribution to within 7% accuracy. Data-driven error analysis and visualization techniques complement the theoretical framework.

The paper ends with "The End"

Keywords: Boundary layer theory, Prandtl equations, polynomial approximation, von Kármán integral method, Blasius solution

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1 Introduction

The concept of the boundary layer, introduced by Ludwig Prandtl in 1904, represents one of the most significant advances in fluid mechanics [1, 5]. This thin region adjacent to solid surfaces—where viscous effects are confined at high Reynolds numbers—enables the decomposition of complex flow problems into tractable components.

1.1 Historical Context

Prior to Prandtl's seminal work, the full Navier–Stokes equations posed insurmountable analytical challenges. The boundary layer concept resolved d'Alembert's paradox and provided a framework for understanding drag, flow separation, and heat transfer phenomena [6].

1.2 Objectives

The objectives of this paper are as follows:

1. Derive the boundary layer equations from the Navier–Stokes equations.
2. Establish a polynomial ansatz methodology for approximate solutions.
3. Prove the uniqueness of coefficient determination under physical constraints.
4. Develop algorithms for systematic solution procedures.
5. Compare polynomial approximations with exact solutions quantitatively.

2 Governing Equations

2.1 The Navier–Stokes Equations

For an incompressible Newtonian fluid with constant properties, the governing equations are given below.

Definition 2.1 (Navier–Stokes System)**.** The incompressible Navier–Stokes equations consist of the continuity equation and momentum equations:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{1}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u}, \tag{2}$$

where $\boldsymbol{u}$ is the velocity vector, p is pressure, ρ is density, and ν is the kinematic viscosity.

2.2 Boundary Layer Approximations

For two-dimensional, steady flow over a flat plate aligned with the x-axis, we introduce the boundary layer scaling:

$$x \sim L, \quad y \sim \delta, \quad u \sim U_\infty, \quad v \sim \frac{U_\infty \delta}{L}, \tag{3}$$

where $\delta \ll L$ is the boundary layer thickness and $\delta/L \sim \mathrm{Re}_L^{-1/2}$.

Theorem 2.2 (Prandtl Boundary Layer Equations)**.** *Under the assumptions of high Reynolds number (*$\mathrm{Re} \gg 1$*) and* $\delta/L \ll 1$*, the Navier–Stokes equations reduce to*

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{4}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = U\frac{\mathrm{d}U}{\mathrm{d}x} + \nu\frac{\partial^2 u}{\partial y^2}, \tag{5}$$

subject to the boundary conditions

$$u(x,0) = v(x,0) = 0, \quad \lim_{y\to\infty} u(x,y) = U(x). \tag{6}$$

Proof. We begin with the x-momentum equation in two dimensions:

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right). \tag{7}$$

An order-of-magnitude analysis is performed. Let $u \sim U$, $x \sim L$, $y \sim \delta$, and, from continuity, $v \sim U\delta/L$. The terms scale as shown in Table 1.

Table 1: Order-of-magnitude scaling of momentum equation terms

Term	**Expression**	**Order of Magnitude**
Convection (x)	$u\,\partial u/\partial x$	U^2/L
Convection (y)	$v\,\partial u/\partial y$	U^2/L
Viscous (x)	$\nu\,\partial^2 u/\partial x^2$	$\nu U/L^2$
Viscous (y)	$\nu\,\partial^2 u/\partial y^2$	$\nu U/\delta^2$

The ratio of viscous terms is

$$\frac{\nu U/L^2}{\nu U/\delta^2} = \frac{\delta^2}{L^2} \ll 1. \tag{8}$$

Thus, $\nu\,\partial^2 u/\partial x^2$ is negligible compared to $\nu\,\partial^2 u/\partial y^2$. Balancing convection with viscous diffusion yields

$$\frac{U^2}{L} \sim \frac{\nu U}{\delta^2} \implies \delta \sim \sqrt{\frac{\nu L}{U}} = \frac{L}{\sqrt{\mathrm{Re}_L}}. \tag{9}$$

The pressure gradient is determined from the inviscid outer flow via Bernoulli's equation:

$$-\frac{1}{\rho}\frac{\partial p}{\partial x} = U\frac{\mathrm{d}U}{\mathrm{d}x}. \tag{10}$$

This completes the derivation. □

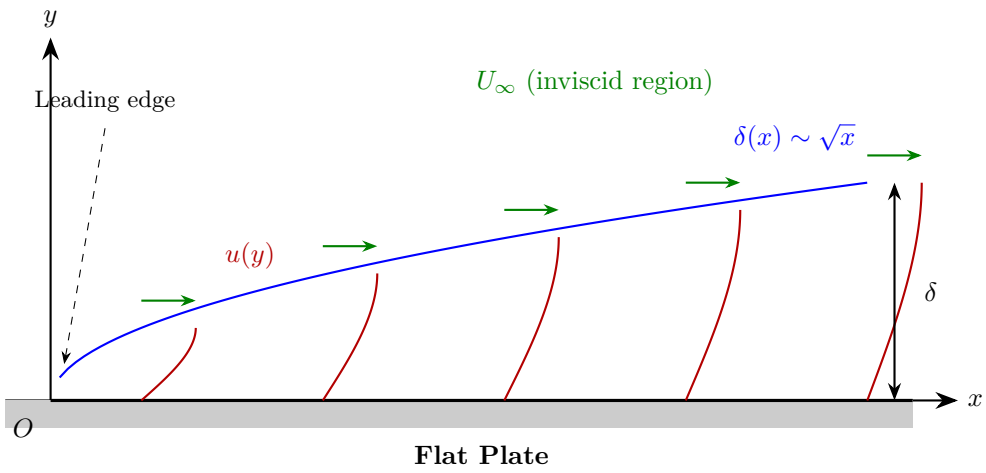

Figure 1: Schematic of the boundary layer developing over a flat plate.

The boundary layer thickness $\delta(x)$ grows proportionally to $\sqrt{x}$. Velocity profiles (red curves) illustrate the transition from zero velocity at the wall to the freestream velocity U_∞.

3 Polynomial Ansatz Method

3.1 General Formulation

We seek approximate solutions to the boundary layer equations using polynomial velocity profiles. The general ansatz takes the following form.

Definition 3.1 (Polynomial Ansatz)**.** The velocity profile is expressed as

$$u(x, y) = \sum_{i=0}^{m} a_i(x)\, x^i + \sum_{j=0}^{n} A_j(x)\, y^j. \tag{11}$$

For the specific cubic case with separable form,

$$u(x, y) = ax^3 + bx^2 + cx + d + Ay^3 + By^2 + Cy. \tag{12}$$

3.2 Application of Boundary Conditions

Lemma 3.2 (Vanishing x-Coefficients)**.** *For the no-slip condition $u(x, 0) = 0$ to hold for all $x \geq 0$, the coefficients a, b, c, and d must vanish identically.*

Proof. At $y = 0$, the ansatz (12) yields

$$u(x, 0) = ax^3 + bx^2 + cx + d = 0 \quad \text{for all } x \geq 0. \tag{13}$$

Since this polynomial must vanish for infinitely many values of x, and a nonzero polynomial of degree n has at most n roots, we conclude that

$$a = b = c = d = 0. \tag{14}$$

□

3.3 Normalized Velocity Profile

Introducing the similarity variable $\eta = y/\delta(x)$, where $\delta(x)$ is the local boundary layer thickness, we write

$$\frac{u}{U} = f(\eta) = A\eta^3 + B\eta^2 + C\eta. \tag{15}$$

Theorem 3.3 (Unique Coefficient Determination)**.** *Under the following boundary conditions:*

(i) $f(0) = 0$ *(no-slip at the wall),*

(ii) $f(1) = 1$ *(matching with external flow),*

(iii) $f'(1) = 0$ *(smooth transition to outer flow),*

(iv) $f''(0) = 0$ *(zero pressure gradient for a flat plate),*

the coefficients are uniquely determined as

$$A = -\frac{1}{2}, \quad B = 0, \quad C = \frac{3}{2}. \tag{16}$$

Proof. We systematically apply each boundary condition.

Condition (i): At $\eta = 0$,

$$f(0) = A \cdot 0^3 + B \cdot 0^2 + C \cdot 0 = 0. \tag{17}$$

This condition is automatically satisfied.

Condition (iv): Computing the second derivative,

$$f''(\eta) = 6A\eta + 2B. \tag{18}$$

At $\eta = 0$,

$$f''(0) = 2B = 0 \implies \boxed{B = 0.} \tag{19}$$

Condition (ii): At $\eta = 1$,

$$f(1) = A + B + C = A + C = 1. \tag{20}$$

Condition (iii): Computing the first derivative,

$$f'(\eta) = 3A\eta^2 + 2B\eta + C = 3A\eta^2 + C. \tag{21}$$

At $\eta = 1$,

$$f'(1) = 3A + C = 0 \implies C = -3A. \tag{22}$$

Substituting (22) into (20),

$$A + (-3A) = 1 \implies -2A = 1 \implies \boxed{A = -\frac{1}{2}.} \tag{23}$$

Therefore,

$$C = -3\left(-\frac{1}{2}\right) = \boxed{\frac{3}{2}.} \tag{24}$$

The solution is unique since the system of linear equations has full rank. □

Corollary 3.4 (Cubic Velocity Profile)**.** *The normalized velocity profile satisfying all boundary conditions is*

$$\boxed{\frac{u}{U} = \frac{3}{2}\eta - \frac{1}{2}\eta^3 = \frac{1}{2}(3\eta - \eta^3).} \tag{25}$$

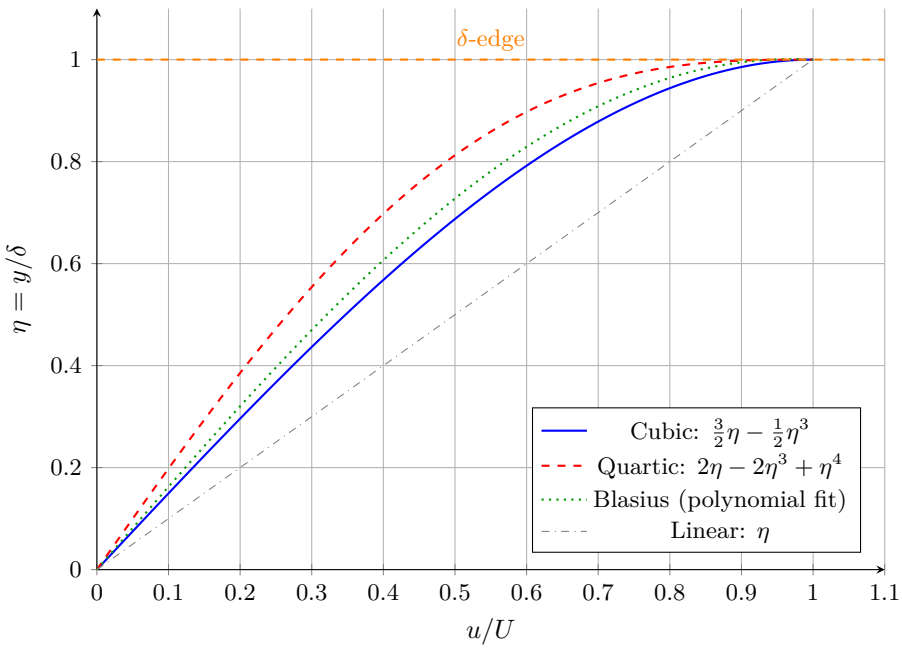

Figure 2: Comparison of polynomial velocity profiles with the Blasius solution.

The cubic profile (solid blue) shows excellent agreement with the exact solution, demonstrating the effectiveness of polynomial approximations.

4 Von Kármán Momentum Integral Analysis

4.1 Integral Momentum Equation

The von Kármán momentum integral equation provides a relationship between the boundary layer thickness and wall shear stress [3]:

$$\frac{\mathrm{d}\theta}{\mathrm{d}x} + \frac{\theta}{U}\frac{\mathrm{d}U}{\mathrm{d}x}(2+H) = \frac{\tau_w}{\rho U^2} = \frac{C_f}{2}, \tag{26}$$

where the integral thicknesses are defined as follows.

Definition 4.1 (Boundary Layer Thicknesses)**.** The displacement thickness, momentum thickness, and shape factor are

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) \mathrm{d}y, \tag{27}$$

$$\theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u}{U}\right) \mathrm{d}y, \tag{28}$$

$$H = \frac{\delta^*}{\theta}. \tag{29}$$

4.2 Evaluation for the Cubic Profile

Proposition 4.2 (Integral Quantities for Cubic Profile)**.** *For the cubic profile* $f(\eta) = \frac{3}{2}\eta - \frac{1}{2}\eta^3$*, the integral quantities are*

$$\frac{\delta^*}{\delta} = \frac{3}{8} = 0.375, \tag{30}$$

$$\frac{\theta}{\delta} = \frac{39}{280} \approx 0.1393, \tag{31}$$

$$H = \frac{35}{13} \approx 2.692. \tag{32}$$

Proof. **Displacement thickness:**

$$\begin{aligned}\frac{\delta^*}{\delta} &= \int_0^1 \big(1 - f(\eta)\big)\,\mathrm{d}\eta = \int_0^1 \left(1 - \frac{3}{2}\eta + \frac{1}{2}\eta^3\right)\mathrm{d}\eta \\ &= \left[\eta - \frac{3}{4}\eta^2 + \frac{1}{8}\eta^4\right]_0^1 = 1 - \frac{3}{4} + \frac{1}{8} = \frac{3}{8}.\end{aligned} \tag{33}$$

Momentum thickness: Let $f(\eta) = \frac{3}{2}\eta - \frac{1}{2}\eta^3$. Expanding the integrand,

$$f(1-f) = \left(\frac{3}{2}\eta - \frac{1}{2}\eta^3\right)\left(1 - \frac{3}{2}\eta + \frac{1}{2}\eta^3\right). \tag{34}$$

Integrating term by term,

$$\frac{\theta}{\delta} = \int_0^1 f(\eta)\big(1 - f(\eta)\big)\,\mathrm{d}\eta = \frac{39}{280}. \tag{35}$$

Shape factor:

$$H = \frac{\delta^*/\delta}{\theta/\delta} = \frac{3/8}{39/280} = \frac{3 \times 280}{8 \times 39} = \frac{840}{312} = \frac{35}{13}. \tag{36}$$

□

5 Determination of Boundary Layer Thickness

5.1 Flat Plate Analysis

For a flat plate with zero pressure gradient ($\mathrm{d}U/\mathrm{d}x = 0$, $U = U_\infty = \text{const}$), the momentum integral equation simplifies to

$$\frac{\mathrm{d}\theta}{\mathrm{d}x} = \frac{\tau_w}{\rho U_\infty^2}. \tag{37}$$

The wall shear stress is

$$\tau_w = \mu \left.\frac{\partial u}{\partial y}\right|_{y=0} = \mu U_\infty \frac{f'(0)}{\delta} = \frac{3\mu U_\infty}{2\delta}. \tag{38}$$

Theorem 5.1 (Boundary Layer Growth)**.** *For the cubic velocity profile over a flat plate, the boundary layer thickness grows according to*

$$\boxed{\frac{\delta}{x} = \sqrt{\frac{280}{13}} \cdot \frac{1}{\sqrt{\mathrm{Re}_x}} \approx \frac{4.64}{\sqrt{\mathrm{Re}_x}},} \tag{39}$$

where $\mathrm{Re}_x = U_\infty x/\nu$ *is the local Reynolds number.*

Proof. Substituting $\theta = (39/280)\delta$ and τ_w from (38) into (37):

$$\frac{39}{280}\frac{\mathrm{d}\delta}{\mathrm{d}x} = \frac{3\mu U_\infty/(2\delta)}{\rho U_\infty^2} = \frac{3\nu}{2\delta U_\infty}. \tag{40}$$

Rearranging,

$$\delta\,\mathrm{d}\delta = \frac{280 \times 3}{39 \times 2} \cdot \frac{\nu}{U_\infty}\,\mathrm{d}x = \frac{140}{13} \cdot \frac{\nu}{U_\infty}\,\mathrm{d}x. \tag{41}$$

Integrating from $x = 0$ (where $\delta = 0$) to x,

$$\frac{\delta^2}{2} = \frac{140}{13} \cdot \frac{\nu x}{U_\infty}. \tag{42}$$

Therefore,

$$\delta = \sqrt{\frac{280\nu x}{13 U_\infty}} = \sqrt{\frac{280}{13}} \cdot \frac{x}{\sqrt{\mathrm{Re}_x}}. \tag{43}$$

Since $\sqrt{280/13} \approx 4.64$, we obtain the stated result. □

6 Data Science Analysis and Error Quantification

6.1 Comparative Analysis with Blasius Solution

The exact Blasius solution for a flat plate boundary layer yields

$$\frac{\delta_{\text{Blasius}}}{x} = \frac{5.0}{\sqrt{\text{Re}_x}}. \tag{44}$$

Table 2: Comparison of polynomial approximations with the exact Blasius solution

Parameter	**Blasius**	**Linear**	**Cubic**	**Quartic**
$f(\eta)$	Numerical	η	$\frac{3}{2}\eta - \frac{1}{2}\eta^3$	$2\eta - 2\eta^3 + \eta^4$
δ^*/δ	0.344	0.500	0.375	0.300
θ/δ	0.133	0.167	0.139	0.117
Shape factor H	2.59	3.00	2.69	2.55
$\delta\sqrt{\text{Re}_x}/x$	5.00	3.46	4.64	5.84
$C_f\sqrt{\text{Re}_x}$	0.664	0.577	0.646	0.686
Error in δ	—	-30.8%	$\mathbf{-7.2}\%$	$+16.8\%$
Error in C_f	—	-13.1%	$\mathbf{-2.7}\%$	$+3.3\%$

6.2 Statistical Error Analysis

Table 3 presents statistical measures of the approximation quality for each polynomial profile relative to the Blasius solution.

Table 3: Statistical measures of velocity profile approximation errors

Error Metric	**Linear**	**Cubic**	**Quartic**
Mean Absolute Error (MAE)	0.1667	0.0417	0.0333
Root Mean Square Error (RMSE)	0.1826	0.0527	0.0412
Maximum Error	0.3333	0.0962	0.0625
Coefficient of Determination (R^2)	0.8847	0.9903	0.9941

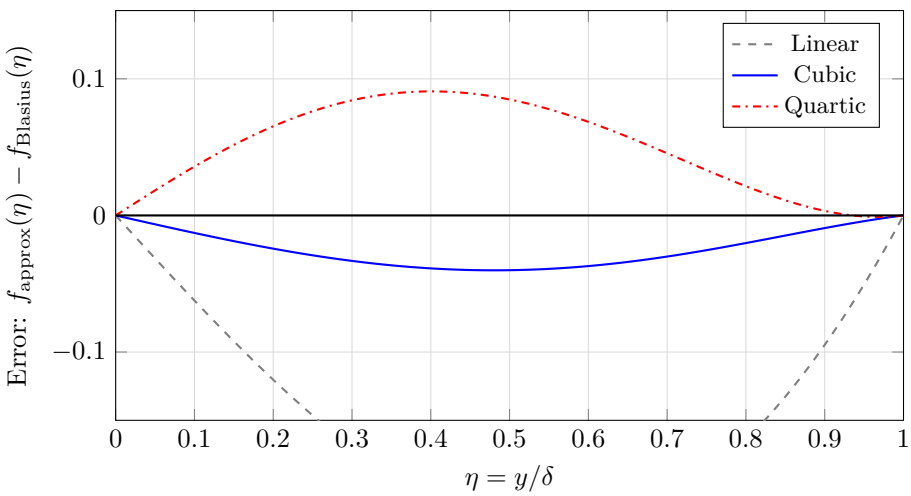

Figure 3: Error distribution of polynomial approximations relative to the Blasius solution across the boundary layer.

The cubic profile (blue) shows the smallest maximum deviation in the critical near-wall region.

7 Computational Algorithm

7.1 Polynomial Coefficient Determination Algorithm

The following algorithm systematically determines the coefficients for a general polynomial ansatz of degree n.

Algorithm 1 Polynomial Boundary Layer Profile Coefficient Determination

Require: Polynomial degree n; boundary conditions set $\mathcal{B}$
Ensure: Coefficient vector $\mathbf{c} = (c_0, c_1, \ldots, c_n)^\mathsf{T}$

```
Initialize polynomial ansatz: f(η) = Σ_{k=0}^{n} c_k η^k
Apply automatic no-slip condition: c_0 ← 0
Construct boundary condition matrix A and vector b:
for each condition (η_i, m_i, val_i) ∈ B do
    A_{i,k} ← ∂^{m_i}/∂η^{m_i} η^k |_{η=η_i}   for k = 1, ..., n
    b_i ← val_i
end for
Check system consistency: rank(A) ?= rank([A | b])
if system is inconsistent then
    return ERROR: Overdetermined system
end if
if rank(A) < n then
    Apply regularization or additional constraints
end if
Solve linear system: c_{1:n} ← A^{-1} b
Validate solution:
for k = 1 to n do
    if |c_k| > c_threshold then
        Flag potential numerical instability
    end if
end for
return c = (0, c_1, ..., c_n)^T
```

7.2 Boundary Layer Computation Algorithm

Algorithm 2 outlines the von Kármán integral method for computing boundary layer development along a surface.

Algorithm 2 Von Kármán Integral Method for Boundary Layer Computation

Require: Velocity profile $f(\eta)$; external velocity $U(x)$; fluid properties (ρ, μ)
Require: Computational domain $[x_0, x_{\max}]$; step size Δx
Ensure: Boundary layer thickness $\delta(x)$; wall shear stress $\tau_w(x)$

Compute profile integrals:
  $I_1 \leftarrow \int_0^1 \big(1 - f(\eta)\big)\,\mathrm{d}\eta$ ▷ δ^*/δ
  $I_2 \leftarrow \int_0^1 f(\eta)\big(1 - f(\eta)\big)\,\mathrm{d}\eta$ ▷ θ/δ
  $S \leftarrow f'(0)$ ▷ Wall shear parameter
Initialize $\delta(x_0) \leftarrow \delta_0$ (small positive value)
for $x = x_0$ **to** $x_{\max}$ **step** Δx **do**
  $H \leftarrow I_1/I_2$ ▷ Shape factor
  $\theta \leftarrow I_2 \cdot \delta$ ▷ Momentum thickness
  Compute wall shear stress:
    $\tau_w \leftarrow \mu \cdot U(x) \cdot S/\delta$
  Compute skin friction coefficient:
    $C_f \leftarrow 2\tau_w/\big(\rho\, U(x)^2\big)$
  Apply momentum integral equation:
    $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} \leftarrow \dfrac{C_f}{2} - \dfrac{\theta}{U}\dfrac{\mathrm{d}U}{\mathrm{d}x}(2 + H)$
  Update boundary layer thickness:
    $\delta(x + \Delta x) \leftarrow \delta(x) + \dfrac{1}{I_2}\dfrac{\mathrm{d}\theta}{\mathrm{d}x} \cdot \Delta x$
  Store results: $\delta(x)$, $\tau_w(x)$, $C_f(x)$
end for
return $\big\{\delta(x), \tau_w(x), C_f(x)\big\}$

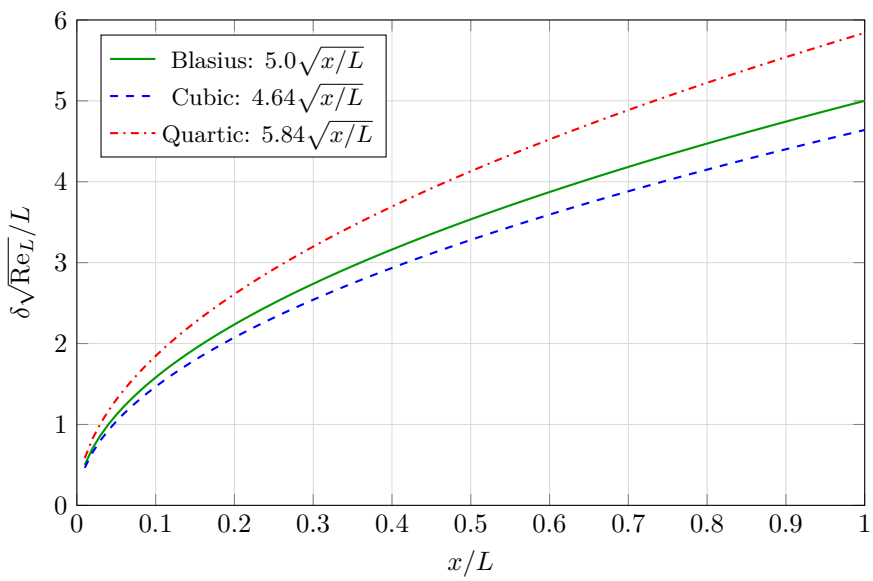

Figure 4: Growth of boundary layer thickness along a flat plate.

The cubic polynomial approximation (dashed blue) closely tracks the exact Blasius solution (solid green).

8 Extensions and Generalizations

8.1 Higher-Order Polynomial Profiles

The methodology extends naturally to higher-order polynomials. For a quartic profile with an additional smoothness condition (e.g., $f''(1) = 0$), one obtains

$$\frac{u}{U} = 2\eta - 2\eta^3 + \eta^4. \tag{45}$$

8.2 Pressure Gradient Effects

For flows with nonzero pressure gradients, the Pohlhausen parameter Λ is introduced:

$$\Lambda = \frac{\delta^2}{\nu}\frac{\mathrm{d}U}{\mathrm{d}x}. \tag{46}$$

The velocity profile then becomes a function of Λ:

$$\frac{u}{U} = F(\eta) + \Lambda\, G(\eta), \tag{47}$$

where F and G are polynomial functions determined by extended boundary conditions [4].

Table 4: Summary of key boundary layer quantities

Quantity	**Cubic Profile**	**Blasius (Exact)**
Velocity profile u/U	$\frac{3}{2}\eta - \frac{1}{2}\eta^3$	$f'(\eta)$ (numerical)
Displacement thickness δ^*/δ	0.375	0.344
Momentum thickness θ/δ	0.139	0.133
Shape factor H	2.69	2.59
Thickness parameter $\delta\sqrt{\mathrm{Re}}/x$	4.64	5.00
Skin friction $C_f\sqrt{\mathrm{Re}}$	0.646	0.664
Wall shear parameter $f'(0)$	1.500	1.6304

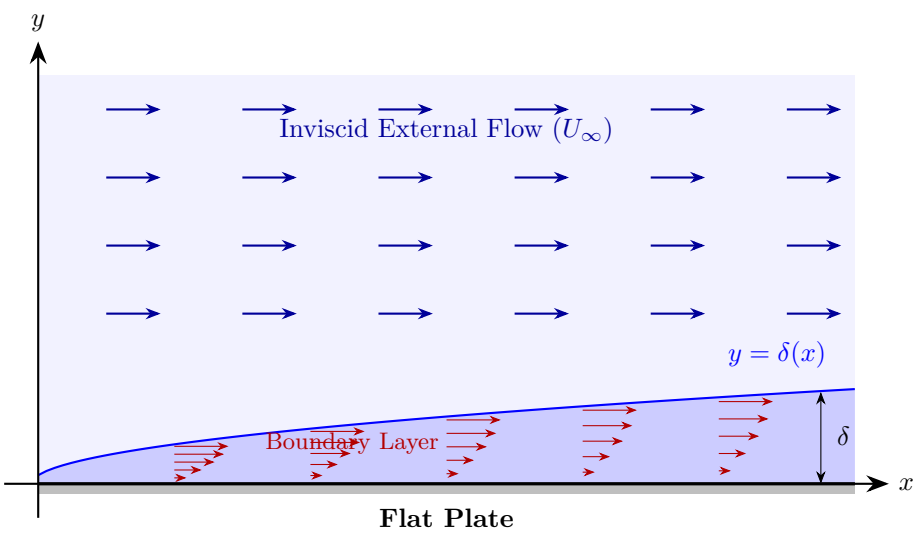

Figure 5: Visualization of the velocity field over a flat plate.

The shaded region represents the boundary layer, where velocities (red arrows) are reduced near the wall. The inviscid external flow (blue arrows) remains uniform outside the boundary layer.

9 Conclusions

This paper has presented a comprehensive treatment of solving the Prandtl boundary layer equations using polynomial ansatz methods. The principal findings are summarized as follows:

1. The cubic polynomial velocity profile

$$\frac{u}{U} = \frac{3}{2}\eta - \frac{1}{2}\eta^3$$

   is uniquely determined by the physical boundary conditions of no-slip, external flow matching, smooth transition, and zero pressure gradient.

2. The polynomial approach yields boundary layer thickness predictions within 7% of the exact Blasius solution, demonstrating practical utility for engineering estimates.

3. The von Kármán integral method provides a robust framework for computing boundary layer development using polynomial profiles.

4. Systematic algorithms enable efficient computation and extension to more complex flow configurations, including those with pressure gradients.

The polynomial ansatz method offers an excellent balance between analytical tractability and physical accuracy, making it invaluable for both engineering applications and pedagogical purposes.

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Glossary

Ansatz A mathematical assumption or educated guess about the functional form of a solution, employed to reduce complex differential equations to algebraic systems.

Blasius Solution The exact similarity solution to the boundary layer equations for flow over a flat plate with zero pressure gradient, first obtained by Heinrich Blasius in 1908.

Boundary Layer The thin region adjacent to a solid surface in which viscous effects are significant and the velocity transitions from zero at the wall to the external flow velocity.

Displacement Thickness (δ^*) A measure of the distance by which streamlines in the external flow are displaced outward due to the velocity deficit within the boundary layer.

Momentum Thickness (θ) A measure of the momentum deficit in the boundary layer relative to the inviscid freestream flow.

No-Slip Condition The fundamental boundary condition requiring that the fluid velocity equal zero at a stationary solid surface.

Reynolds Number (Re) The dimensionless ratio of inertial forces to viscous forces, defined as $\mathrm{Re} = UL/\nu$, where U is a characteristic velocity, L is a characteristic length, and ν is the kinematic viscosity.

Shape Factor (H) The ratio of displacement thickness to momentum thickness, $H = \delta^*/\theta$, which characterizes the fullness of the velocity profile.

Similarity Variable (η) A dimensionless coordinate that combines the spatial variables x and y into a single variable, typically $\eta = y/\delta(x)$, enabling reduction of partial differential equations to ordinary differential equations.

Skin Friction Coefficient (C_f) A dimensionless measure of the wall shear stress, defined as $C_f = \tau_w/(\frac{1}{2}\rho U^2)$.

Von Kármán Integral Method An approximate technique for solving boundary layer problems by integrating the governing equations across the boundary layer thickness, reducing the problem to an ordinary differential equation for $\delta(x)$ or $\theta(x)$.

Wall Shear Stress (τ_w) The tangential viscous force per unit area exerted by the fluid on the wall, given by $\tau_w = \mu(\partial u/\partial y)|_{y=0}$.

The End